

ARPAN PRAMANIK

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Summary

Computer Science undergraduate specializing in AI/ML with research and software engineering experience across applied machine learning, backend systems, and GenAI. Experienced in developing deployable AI solutions through research, internships, and end-to-end projects, with a strong interest in technology-driven solutions for real-world societal problems.

Education

The Neotia University

B.Tech in Computer Science and Engineering (AIML)

2023 – 2027

CGPA: 9.42

Sundarban Adarsha Vidyamandir

XII (WBCHSE)

2022

Grade: 87.6%

Experience

AI/ML Research Intern

Jun 2026 – Jul 2026

National Institute of Technology (NIT) Durgapur

- Investigated 7 EfficientNet-B0 variants in TensorFlow using quadratic convolution formulations and RBF, Polynomial, Swish, Leaky ReLU, and ELU activations.
- Benchmarked variants against the 78.72% baseline, analyzing gradient behavior, numerical stability, and activation performance to identify the most stable configuration.

Web Development Intern

Jun 2025 – Sep 2025

Xeta Labs, Guwahati, India

- Developed and integrated a production LMS across Student, Tutor, and Admin roles, implementing end-to-end application workflows.
- Debugged and resolved 50+ backend issues before launch involving authentication, REST APIs, and database reliability.

Projects

JalDrishti — Climate-Smart Precision Irrigation Platform

Live Demo | GitHub

- Tech Stack:** Python, FastAPI, Flutter, RAG, LLMs, FAO-56, NASA POWER, SoilGrids.
- Developed a climate-smart irrigation advisory platform combining FAO-56 Penman–Monteith evapotranspiration modeling, satellite-derived soil data, and real-time weather information to generate irrigation recommendations.
- Built a multilingual AI advisory layer using RAG and LLMs to translate agronomic insights into actionable recommendations for farmers.
- Designed the platform to optimize water usage and pump energy consumption while tracking cumulative financial savings and irrigation efficiency.

PaperLens AI — Autonomous RAG Research Assistant

Live Demo | GitHub

- Tech Stack:** React, TypeScript, FastAPI, Supabase (pgvector), FAISS, LLMs, RAG, MCP.
- Built a framework-free Agent Mode for autonomous literature discovery, methodology analysis, research-gap identification, dataset recommendation, and experiment planning.
- Engineered deterministic routing, scoped tool selection, structured outputs, prompt compression, and multi-provider fallbacks to improve reliability and reduce unnecessary LLM calls.
- Integrated persistent vector retrieval, citation validation, request batching, and SSE-based progress streaming for responsive research workflows.

FruitQ-GradeX — Explainable Fruit Quality Assessment

Live Demo | GitHub

- Tech Stack:** TensorFlow, Keras, OpenCV, Streamlit, Grad-CAM, NumPy, Pandas.
- Developed a multi-head CNN to jointly classify 4 fruit types and quality conditions using 9,146 images, achieving 96.8% and 98.5% test accuracy respectively.
- Integrated Grad-CAM for explainable predictions and deployed a Streamlit application supporting real-time image and webcam-based assessment.

Research Publications

- S. Dutta, S. G. Neogi, D. Chanda, **A. Pramanik**, Ö. Girgin and E. L. Öncül, "FruitQ-GradeX: Determining Fruit Quality and Grading with Explainable Deep Learning," *IEEE ICRITO 2025*. DOI
- S. Dutta, S. G. Neogi, S. Chowdhury, V. Chole, **A. Pramanik** and D. Chanda, "Cropsense: Explainable Deep Learning Framework for Accurate Quality Detection in Solanaceous Crops," *IEEE ICRITO 2025*. DOI

Skills

Languages: Python, Java, C, JavaScript, SQL

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

AI/ML: PyTorch, TensorFlow, scikit-learn, OpenCV

GenAI: RAG, LLMs, Vector Search, MCP

Software Engineering: FastAPI, Node.js, Express.js, React.js, REST APIs

Databases & Cloud: PostgreSQL, MongoDB, Supabase, AWS, Redis

DevOps: Docker, Git, GitHub, GitHub Actions, OpenTelemetry, Prometheus